D 30571 **(Pages : 4)** **Name.....**

Reg. No.....

FIFTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION, NOVEMBER 2022

Mathematics

MTS 5B 08—LINEAR PROGRAMMING

(2020 Admission onwards)

Time : Two Hours Maximum : 60 Marks

Section A

Answer any number of questions.
Each question carries 2 *marks.*
Ceiling is 20.

1. Define optimal solution.

2. Draw and shade a bounded polyhedral convex subset in $\mathbb{R}^2$.

3. Represent the following linear programming problem into canonical slack minimization linear programming problem ;

   Minimize $c(x_1, x_2, x_3) = 20x_1 + 30x_2 + 25x_3$ subject to

   $x_1 + 2x_2 + 2x_3 \geq 200$
   $2x_1 + 2x_2 + x_3 \geq 150$
   $x_1, x_2, x_3 \geq 0$

4. Pivot on 5 in the canonical maximum tableau given below :

x_1	x_2	-1		
1	2	3	=	$-t_1$
4	5	6	=	$-t_2$
7	8	9	=	f

5. Write the simplex algorithm for maximum basic feasible tableaus.

6. Define negative transpose.

Turn over

7. Consider the canonical minimization linear programming problem given below :

   Minimize $g(y_1, y_2) = -y_2$ subject to

   $y_1 - y_2 \geq 1$
   $-y_1 + y_2 \geq 2$
   $y_1, y_2 \geq 0.$

   State the dual canonical maximization linear programming problem.

8. If a canonical maximization linear programming problem is unbounded, prove that the dual canonical minimization linear programming problem is infeasible.

9. Define complementary slackness.

10. Explain von Neumann minimax theorem.

11. Explain briefly the minimum entry method to obtain an initial basic feasible solution in transportation problem..

12. What is an assignment problem ?

Section B

Answer any number of questions.
Each question carries 5 *marks.*
Ceiling is 30.

13. Using geometrical method, solve the following linear programming problem :

   Maximize $f(x, y) = 5x + 2y$ subject to

   $x + 3y \leq 14$
   $2x + y \leq 8$
   $x, y \geq 0.$

14. Solve the following canonical linear programming problem by simplex method :

x	y	-1		
-1	-1	-2	=	$-t_1$
1	-2	0	=	$-t_2$
-2	1	1	=	$-t_3$
-1	3	0	=	f

15. Solve the noncanonical linear programming problem given below :

Maximize $f(x,y) = x + 3y$ subject to

$x + 2y \leq 10$
$3x + y \leq 15$
$x \geq 0.$

16. State and prove duality theorem.

17. Find the von Neumann value and the optimal strategy for each players in the matrix game given below :

		II		
	2	1	4	2
	1	2	1	1
I	–2	6	3	–2
	3	–3	5	1
	1	2	2	1

18. Explain briefly north west corner method and use it to obtain the initial basic feasible solution of the transportation problem given below :

2	1	2	40
9	4	7	60
1	2	9	10
40	50	20	

19. Solve the assignment problem given below :

38	21	34
41	14	36
28	20	25

Section C

Answer any **one** *question.*
The question carries 10 *marks.*

20. Solve the dual noncanonical linear programming problem given below :

	x_1	x_2	x_2	–1	
y_1	0	–1	–1	–1	= 0
y_2	–1	–3	4	0	$= -t_1$
y_3	–1	2	–3	0	$= -t_2$
–1	–1	0	0	0	$= f$
	= 0	$= s_1$	$= s_2$	$= g$	

Turn over

21. Write transportation algorithm and use this algorithm solve the following transportation problem :

5	12	8	50	26
11	4	10	8	20
14	50	1	9	30
15	20	26	15	

(1 × 10 = 10 marks)